SQL ANALYSIS & METHODOLOGY REPORT

Query workflow, analytical themes, validation controls, and interpretation guidance

Project period: 2010–2024 **Category:** Business Intelligence & Data Analytics

Tools: MySQL, Tableau Public, Microsoft Excel **Source format:** CSV

1. Purpose of the SQL Layer

The SQL component turns the source CSV data into structured summaries that can be checked, compared, and used to inform the Tableau dashboards. The project's existing script is named BMW_Sales_Analysis.sql, and its query results are organized into themed folders. This report documents the analytical workflow without claiming specific numerical outcomes that have not been supplied for review.

2. Analysis Workflow

Step 1 — Data setup and overview

- Confirm the database/schema and imported table name.
- Inspect the table structure and field data types.
- Record row count, date/year range, and distinct category counts.

Step 2 — Data quality audit

- Check null or blank values in important dimensions and measures.
- Look for duplicate rows or repeated business keys where a key exists.
- Review numeric ranges for price, sales volume, engine size, and mileage.
- Standardize category spelling only when the intended correction is clear and documented.

Step 3 — KPI and business performance analysis

- Summarize the available sales measures overall and by year.
- Calculate averages only where an average is meaningful for the field.
- Use explicit aliases and readable output columns for downstream reporting.

Step 4 — Product and market deep analysis

- Group results by model and other product attributes.

- Compare regional totals and distributions using consistent measures.
- Use ORDER BY and LIMIT for ranked lists, and state the metric used for ranking.

Step 5 — Advanced sales analysis

- Use subqueries or grouped intermediate results where needed.
- Compare a group-level metric with a clearly defined benchmark.
- Check edge cases such as ties, nulls, and groups with few observations.

3. SQL Concepts Demonstrated

The folder structure indicates analysis themes covering setup, data quality, KPI performance, product/market analysis, advanced sales analysis, and regional/model analysis. The main script can demonstrate SELECT, WHERE, GROUP BY, aggregate functions such as SUM/AVG/COUNT/MAX, HAVING, ORDER BY, subqueries, and aliases where these are present in the saved queries. Only list a technique as completed if it appears in the actual script.

4. Metric Definitions and Aggregation Rules

Measure	Potential interpretation	Validation question
Sales volume	Total units or recorded volume, depending on source definition.	Does each row represent a sale, a group of sales, or a summary?
Price (USD)	Listed/recorded vehicle price.	Is it unit price, average price, or another price measure?
Estimated revenue	Price × sales volume only if definitions support this calculation.	Are price and volume aligned at the same row grain and time basis?
Average mileage	Mean of recorded mileage values.	Are units consistently kilometers and are values plausible?
Model/region ranking	Groups ranked by a chosen metric.	Is the ranking metric named and the aggregation consistent?

5. Query Validation Protocol

- Run the query against the intended table and confirm it completes without errors.
- Check the result columns, row count, sort order, and units.
- Manually validate a small number of grouped results against the raw data.
- For rankings, confirm whether ties are possible and whether the output handles them as intended.
- Compare corresponding Tableau KPIs or chart totals against the SQL result.
- Document any exclusions, filters, assumptions, or data corrections.

6. Common Risks to Avoid

- Using SUM for a field that should be averaged or counted.
- Calling price totals ‘revenue’ without verifying the data grain and sales-volume definition.
- Treating missing values as zero without a business rule.
- Comparing groups that have different time coverage or inconsistent filters.
- Reporting a top-performing model or region without naming the measure and time period.

7. Results Log Template

Complete this table from actual saved query outputs before using findings in a portfolio, interview, or executive summary.

Analysis	Metric / period	Verified result	SQL result location
Overall performance	Insert metric and years	Insert verified value	Query Result folder
Model comparison	Insert ranking metric	Insert verified model/rank	Query Result folder
Regional analysis	Insert metric and period	Insert verified region/result	Query Result folder
Vehicle attributes	Insert attribute/metric	Insert verified comparison	Query Result folder

8. Reproducibility and Handover

Keep BMW_Sales_Analysis.sql as the main query script. Preserve result files with descriptive names and maintain the relationship between each result and the query that produced it. If the source file or table name changes, update the setup section and retest all dependent queries. This allows another reviewer to understand the analysis path and reproduce the results.

9. Conclusion

The SQL layer is the analytical foundation of the project. Its quality should be judged not only by the number of queries but by correct aggregation, explicit assumptions, validation against source data, readable outputs, and consistency with Tableau. Add quantified findings only after verifying the saved query results.